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1) HTML Introduction

HTML (HyperText Markup Language) is the standard language for creating webpages. Understanding HTML is fundamental for anyone looking to develop web applications or websites.

Use Case Scenario:

Suppose you are a beginner who wants to create a personal blog. Learning HTML is the first step to building the structure of your blog page.

The basic anatomy of an HTML document includes the following elements:

Doctype Declaration: It specifies the version of HTML used for the document.

HTML Element: It is the root element of the HTML document.

Head Element: It contains the metadata of the document, including the title, keywords, and other information.

Body Element: It contains the content of the web page.

2) HTML Basic

The basic structure of an HTML document includes a document type declaration, and the main elements like `<html>`, `<head>`, and `<body>`.

Use Case Scenario:

If you're creating a new HTML page, understanding the basic layout helps you start correctly without missing any key structure.

HTML Structure:

The `<html>` tag is the starting point for every HTML document. It specifies that the document is an HTML document. The `<html>` tag encloses all the other elements in the document.

Example:

```
<!DOCTYPE html>
<html>
<head>
<title>My Web Page</title>
</head>
<body>
<h1>Welcome to my Web Page</h1>
<p>This is my first paragraph.</p>
</body>
</html>
```

<html> is the parent tag

Doctype Declaration: <!DOCTYPE html>

A doctype or document type declaration is an instruction that tells the web browser about the markup language in which the current page is written. **The Doctype is not an element or tag, it lets the browser know about the version of or standard of HTML or any other markup language that is being used in the document.**

The declaration must be the first line of the HTML document.

HTML Element: <html>

The <html> tag is the root element of the HTML document. It is the container for all the other elements in the document.

Head Element: <head>

The <head> tag contains the metadata of the document, including the title, keywords, and other information. The metadata is not displayed on the web page.

Page title: <title>

The <title> tag specifies the title of the web page. It appears in the browser's title bar and is used by search engines to index the page.

Body Element: <body>

The <body> tag contains the content of the page. It includes text, images & multimedia

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Webpage</title>
  </head>
  <body>
    <h1>Welcome to My Blog</h1>
    <p>This is my first webpage using HTML</p>
  </body>
</html>
```

In the above example, the <head> element contains the <title> element that specifies the title of the web page. The <body> element contains the content of the web page, including a <h1> heading and a <p> paragraph.

3)HTML Elements

HTML elements are the building blocks of HTML pages. An element is defined by a start tag, some content, and an end tag.

Use Case Scenario:

You want to display different types of content like headings, paragraphs, links, and images in your web page.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
    <a href="https://www.example.com">This is a link</a>
  </body>
</html>
```

4) HTML Attributes

Attributes provide additional information about HTML elements. They are always specified in the start tag and usually come in name/value pairs like **name="value"**.

Use Case Scenario:

You want to add a tooltip to an image or set the link to open in a new tab. Attributes help define this behavior.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <p title="This is a tooltip">Hover over this paragraph.</p>
    <a href="https://www.example.com" target="_blank">Visit Example</a>
  </body>
</html>
```

5) HTML Headings

HTML headings are used to define headings on a webpage. They range from <h1> (most important) to <h6> (least important).

Use Case Scenario:

When organizing content on a blog, you use headings to structure sections like title, subtitle, and smaller topics.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h1>Main Title</h1>
    <h2>Subtitle</h2>
    <h3>Section Heading</h3>
    <h4>Sub-section</h4>
    <h5>Minor Heading</h5>
    <h6>Smallest Heading</h6>
  </body>
</html>
```

<h1> has largest size & <h6> smaller than all other

6) HTML Paragraphs

HTML paragraphs are defined using the <p> tag. Paragraphs are used to structure text content on the web.

Use Case Scenario:

You want to write a welcome message or article content on your site. Paragraphs help organize the text into readable blocks.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <p>This is the first paragraph.</p>
    <p>This is the second paragraph with more details.</p>
  </body>
</html>
```

7) HTML Styles

HTML styles allow you to add visual styling to elements, like color, font, or size, using the style attribute or CSS.

Use Case Scenario:

You want to change the color and font size of a heading without creating a separate stylesheet.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h1 style="color:blue;">Blue Heading</h1>
    <p style="font-size:18px;">This is a styled paragraph.</p>
  </body>
</html>
```

8) HTML Formatting

Formatting tags are used to format the appearance of text, such as bold, italic, underline, etc.

Use Case Scenario:

When writing a blog, you want to emphasize important words or show code snippets.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <b>Bold text</b><br>
    <i>Italic text</i><br>
    <u>Underlined text</u><br>
    <code>console.log("Hello")</code>
  </body>
</html>
```

9) HTML Quotation and Citation

HTML provides special tags for quoting text and citing references. These include:

<blockquote> – for long quotations.

<q> – for short inline quotes.

<cite> – for citing a source (e.g., book or article title).

<abbr> – for abbreviations.

<address> – for contact information.

Use Case Scenario:

You're writing a blog or article and want to properly quote authors, books, or include citations and abbreviations.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Quotation and Citation</h2>
    <p>Here is a blockquote:</p>
    <blockquote cite="https://www.goodreads.com">
      "It is our choices, Harry, that show what we truly are, far more than our abilities."
    </blockquote>
    <p>Inline quote: <q>Knowledge is power.</q></p>
    <p><abbr title="HyperText Markup Language">HTML</abbr> is used to create webpages.</p>
    <p><cite>Harry Potter and the Chamber of Secrets</cite> by J.K. Rowling</p>
    <address>Written by John Doe<br>john@example.com</address>
  </body>
</html>
```

10) HTML Comments

Comments in HTML help you write notes or explanations that are not visible in the browser. They are written using **<!-- comment -->**.

Use Case Scenario:

You want to leave reminders or hide code for testing without deleting it.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <!-- This is a comment -->
    <p>This is a visible paragraph.</p>
    <!-- <p>This paragraph is commented out and will not show.</p> -->
  </body>
</html>
```


11) HTML Colors

HTML supports colors in:

- Named colors (e.g., red, blue)
- Hex codes (e.g., #ff0000)
- RGB values (e.g., rgb(255,0,0))
- RGBA, HSL, HSLA formats

Use Case Scenario:

You want to apply color to text, background, borders, etc., for styling your webpage.

Example:

```
<!DOCTYPE html>
<html>
  <body style="background-color: lightyellow;">
    <h1 style="color: navy;">This is a navy heading</h1>
    <p style="color: #ff6347;">This text is tomato colored using HEX</p>
    <p style="color: rgb(60, 179, 113);">This is medium sea green text using RGB</p>
  </body>
</html>
```

12) HTML CSS

CSS (Cascading Style Sheets) is used to control layout and appearance. It can be included in:

- Inline (style="")
- Internal (<style> inside <head>)
- External (.css file)

Use Case Scenario:

You want to separate structure from design for better maintainability and cleaner HTML.

Example (Internal CSS):

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      body {
        background-color: #f0f0f0;
      }
      h1 {
        color: darkred;
      }
      p {
        font-family: Arial;
        font-size: 18px;
      }
    </style>
  </head>
  <body>
    <h1>Hello World</h1>
    <p>This is a paragraph</p>
  </body>
</html>
```

<style> tag is going to be used

```
</head>
<body>
  <h1>Styled with Internal CSS</h1>
  <p>This paragraph uses a custom font and size.</p>
</body>
</html>
```

13) HTML Links

HTML links are created using the <a> anchor tag. They require **the href attribute** to specify the destination URL.

Use Case Scenario:

You want to link to another webpage, website, or section of the same page.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <a href="https://www.google.com" target="_blank">Go to Google</a>
    <br>
    <a href="#section2">Jump to Section 2</a>

    <h2 id="section2">Section 2</h2>
    <p>This is the target section.</p>
  </body>
</html>
```

14) HTML Images

The tag is used to embed images. It includes attributes like src, alt, width, and height.

Use Case Scenario:

You want to display logos, profile pictures, or decorative images on your website.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Image Example</h2>
    
  </body>
</html>
```

15) HTML Image Map

Image maps allow multiple clickable areas in a single image using **<map>** and **<area>** tags.

Use Case Scenario:

You want different parts of an image (like a map) to link to different pages.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Image Map Example</h2>
    

    <map name="workmap">
      <area shape="rect" coords="34,44,270,350" alt="Computer" href="computer.html">
      <area shape="rect" coords="290,172,333,250" alt="Phone" href="phone.html">
    </map>
  </body>
</html>
```

16) HTML Background Image

You can set a background image for an HTML element using **CSS background-image property**.

Use Case Scenario:

You want to add a full-page background image to enhance your webpage's look.

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      body {
        background-image: url('https://www.w3schools.com/html/img_girl.jpg');
        background-size: cover;
        background-repeat: no-repeat;
        background-position: center;
      }
    </style>
  </head>
  <body>
    <h1>This page has a background image</h1>
    <p>Using CSS, the background image covers the entire page.</p>
  </body>
</html>
```

17) HTML Favicon

A favicon is a small image displayed next to the page title in the browser tab. It helps with branding and enhances user experience.

Scenario: You are designing a website for a company, and you want their logo to appear in the browser tab. Adding a favicon makes it look professional and helps users recognize the site.

Next, add the **<link>** element (highlighted below) to your index.html file right below the **<title>** element.

Example :- open any code editor like notepad, vs code & create a new file named as index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Page</title>
  <link rel="icon" type="image/x-icon" href="C:\Users\HP\Desktop//OIP.jpg">
</head>
<body>
  <h1>Welcome to My Page</h1>
</body>
</html>
```

Output:- Save the file and reload it in your web browser. Your browser tab should now contain a favicon image.



Supported File Formats :-

The following formats are supported across all the major browsers.

.ico .png .gif .jpg .svg

18) HTML Title

The **<title>** element defines the title of the HTML document. It appears in the browser tab, is used by search engines, and is also shown in bookmarks. **The <title> tag must be placed inside the <head> section** of the HTML document and should be unique and descriptive for every page.

Use Case Scenario:

You are building a website and want to give each page a clear, SEO-friendly title that helps users and search engines identify the content of the page.

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Personal Portfolio</title>
  </head>
  <body>
    <h1>Welcome to My Website</h1>
    <p>This is my personal portfolio where I share my projects and blogs.</p>
  </body>
</html>
```

Explanation:

- In this example, the browser tab will show: "My Personal Portfolio"
- When someone bookmarks this page, the same title will be saved.
- Search engines like Google also use the title as the main heading in search results.

19) Html Tables

A table is a structured set of data made up of rows and columns (tabular data). A table allows you to quickly and easily look up values that indicate some kind of connection between different types of data, for example a person and their age, or a day of the week, or the timetable for a local swimming pool.

To create a table in HTML you will need to use tags. The most important one is the **<table>** tag which is the main container of the table. It shows where the table will begin and where it ends.

Common HTML Table tags

<tr> - represents rows

<td> - used to create data cells

<th> - used to add table headings

<caption> - used to insert captions

<thead> - adds a separate header to the table

<tbody> - shows the main body of the table

<tfoot> - creates a separate footer for the table

Example 1:

Suppose we have this particular table we want to create its structure:

s.no	name	rollno
1	python	101
2	php	102

Code:-

```
<!DOCTYPE HTML>
<html>
<head><title>my tables</title>
</head>
<body>
<table>
<tr>
<th>S.no</th>
<th>Name </th>
<th>Rollno</th>
</tr>
<tr>
<td>1</td>
<td>python</td>
<td>101</td>
</tr>
<tr>
<td>2</td>
<td>php</td>
<td>102</td>
</tr>
</table></body></html>
```

Explanation:-

First we would use <table tag> inside the <body> tag then to start the first row we have <tr> tag. After that <th> to specify the heading in the first row. when first row gets completed then we would close the </tr> tag. Again for next row we start <tr> tag and then <td> to specify the table headings and so on.

➔ **Spanning the rows and columns:**

ROWSPAN :- To merge the rows we use rowspan tag.

Syntax:- <table rowspan ="no of rows we want to merge">

COLSPAN:- To merge the columns we use colspan tag.

Syntax:- <table colspan ="no of columns to be merged">

Example 2:-

```
<!DOCTYPE html>
<html>
<head>
<title>my first page</title>
</head>
<body>
<table border="2">
<tr>
<th>s.no</th>
<th>Name</th>
<th>Rollno</th>
</tr>
<tr>
<td>1</td>
<td rowspan="2">python</td>
<td>101</td>
</tr>
<tr>
<td>2</td>
<td>102</td>
</tr></table>
</body></html>
```

Explanation:

The above example is based on the rowspan i.e., merging the rows

Example 3:-

```
<!DOCTYPE html>
<html>
<head>
<title>my first page</title>
</head>
<body>
<table border="2">
<tr>
<th>Name</th>
<th>Rollno</th>
<th>Marks</th>
</tr>
<tr>
<td>python</td>
<td colspan="2" align="center">90</td>
</tr>
<tr>
<td>php</td>
<td>1</td>
<td>100</td>
</tr></table>
</body></html>
```

Explanation: This example shows how we use merge the column on the basis of our needs.

➔ How to Add a Caption to a Table

The main use of adding a caption to table is to provide a description about the data represented in the table. The caption can either be placed at the top of the table or the bottom and by default it will always be centered.

To insert a caption into a table, use the <caption> tag.

```
<table border="2">
<caption>Welcome to Myanatomy</caption>
<tr>
<th>Sites</th>
<th>courses</th>
<th>services</th>
</tr>
<tr>
<td>Admin Block </td>
<td>Comp sc</td>
<td>all</td>
</tr></table>
```


➔ How to Add a Table Header, Body, & Footer in HTML Tables

Just like how a website or any other document has **three main sections** – the **header, body, and footer** – so does a table.

In a table they are divided by using attributes namely:

<thead> - provides a separate header for the table

<tbody> - contains main content of the table

<tfoot> - creates a separate footer for the table

Example:- with use of <thead>, <tbody> & <tfoot>

<pre><table> <thead> <tr> <th colspan="2">October</th> <th colspan="2">November</th> </tr> </thead> <tbody> <tr> <td>Sales</td> <td>Profit</td> <td>Sales</td> <td>Profit</td> </tr> <tr> <td>\$200,00</td> <td>\$50,00</td> <td>\$300,000</td> <td>\$70,000</td> </tr> </tbody> <tfoot> <tr> <th colspan= "4">November was more produstive</th> </tr> </tfoot> </table></pre>	<pre><THEAD></pre>
<pre><tbody> <tr> <td>Sales</td> <td>Profit</td> <td>Sales</td> <td>Profit</td> </tr> <tr> <td>\$200,00</td> <td>\$50,00</td> <td>\$300,000</td> <td>\$70,000</td> </tr> </tbody></pre>	<pre><TBODY></pre>
<pre><tfoot> <tr> <th colspan= "4">November was more produstive</th> </tr> </tfoot></pre>	<pre><TFOOT></pre>

Explanation:- <thead> <tbody> <tfoot> works in background so that search engine and web browser can identify which one is header part, body content and footer of the table. There will not be any difference in the output after using these tags. These are just for browsers.

One way to style the table

```
<table>
<tr>
<th>Data 1</th>
<th style="background-color: yellow">Data 2</th>
</tr>
<tr>
<td>Calcutta</td>
<td style="background-color: yellow">Orange</td>
</tr>
<tr>
<td>Robots</td>
<td style="background-color: yellow">Jazz</td>
</tr>
</table>
```

Using <col>

Instead of doing this, we can specify the information once, on a **<col>** element. **<col>** elements are specified inside a **<colgroup>** container just below the opening **<table>** tag. We could create the same effect as we see above by specifying our table as follows:

```
<table>
<colgroup>
<col />
<col style="background-color: yellow" />
</colgroup>
<tr>
<th>Data 1</th>
<th>Data 2</th>
</tr>
<tr>
<td>Calcutta</td>
<td>Orange</td>
</tr>
<tr>
<td>Robots</td>
<td>Jazz</td>
</tr>
</table>
```

20) HTML Lists

HTML supports three types of lists:

- **Ordered List ()** – Items are numbered.
- **Unordered List ()** – Items are bulleted.
- **Description List (<dl>)** – Used for name/value pairs.

Use Case Scenario:

You want to organize content like steps in a tutorial, a shopping list, or glossary definitions.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Ordered List</h2>
    <ol>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ol>

    <h2>Unordered List</h2>
    <ul>
      <li>Apple</li>
      <li>Banana</li>
      <li>Cherry</li>
    </ul>

    <h2>Description List</h2>
    <dl>
      <dt>HTML</dt>
      <dd>A markup language for web pages.</dd>
      <dt>CSS</dt>
      <dd>Styles the appearance of web content.</dd>
    </dl>
  </body>
</html>
```

**ordered list **

**unordered list **

description list <dl>

21) HTML Block and Inline Elements

Block-level elements take up the full width and always start on a new line.

Inline elements do not start on a new line and only take as much width as needed.

Use Case Scenario:

Understanding how elements behave in layout helps you structure your pages correctly.

Example:

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Block Elements</h2>
    <div style="background-color: lightblue;">This is a div block</div>
    <p>This is a paragraph (block element).</p>

    <h2>Inline Elements</h2>
    <span style="color: red;">This is a span inside a paragraph.</span>
    <strong>This is bold (inline).</strong>
  </body>
</html>
```

22) HTML Classes

The class attribute is used to define styles for multiple elements. Classes allow reusability and styling with CSS or JavaScript.

Use Case Scenario:

You want to apply the same styling to multiple sections, like giving all buttons the same color or all product cards the same layout.

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      .highlight {
        background-color: yellow;
        font-weight: bold;
      }
    </style>
  </head>
  <body>
    <p class="highlight">This paragraph is highlighted.</p>
    <p>This is normal text.</p>
    <div class="highlight">This div is also highlighted.</div>
  </body>
</html>
```

23) HTML ID

The id attribute uniquely identifies a single HTML element. It can be used for styling, linking (anchors), or JavaScript manipulation.

Use Case Scenario:

You want to create a "back to top" link, or apply specific styles or actions to only one element.

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      #special {
        color: green;
        font-size: 22px;
      }
    </style>
  </head>
  <body>
    <h2 id="special">This heading has a unique style using ID</h2>
    <p>This is a normal paragraph.</p>
    <a href="#special">Go to special heading</a>
  </body>
</html>
```

24) HTML Iframes

An HTML iframe is used to display a web page within a web page.

The HTML <iframe> tag specifies an inline frame.

An inline frame is used to embed another document within the current HTML document.

Scenario: You want to embed a Google Map on your contact page to help visitors locate your business.

It is a good practice to always include a title attribute for the <iframe>. This is used by screen readers to read out what the content of the iframe is.

Example:

```
<!DOCTYPE html>
<html>
<body>
<h2>HTML Iframes</h2>
<p>You can use the height and width attributes to specify the size of the iframe:</p>

<iframe src="demo_iframe.html" height="200" width="300" title="Iframe Example"></iframe>

</body>
</html>
```

Iframe - Target for a Link

An iframe can be used as the target frame for a link.

The target attribute of the link must refer to the name attribute of the iframe.

Example :

```
<!DOCTYPE html><html><body>

<h2>Iframe - Target for a Link</h2>

<iframe src="demo_iframe.html" name="iframe_a" height="300px" width="100%" title="Iframe
Example"></iframe>

<p><a href="https://www.google.com" target="iframe_a">Google</a></p>

</body></html>
```

25) HTML JavaScript

HTML allows embedding JavaScript to make web pages interactive. JavaScript is used for tasks like form validation, animations, event handling, and dynamic content updates.

Scenario: You want a button on your website that, when clicked, displays a greeting message.

Use Case:

- Validating user input in forms.
- Creating interactive UI elements like modals and dropdowns.
- Dynamically updating content without reloading the page.

Example:

```
<!DOCTYPE html>

<html><head>

  <title>JavaScript Example</title>

  <script>
    function showMessage() {
      alert("Hello, World!");
    }
  </script>
</head>

<body>

  <button onclick="showMessage()">Click Me</button>

</body></html>
```

26) HTML File Paths

HTML file paths define the location of files such as images, CSS, and JavaScript within a project structure.

Scenario: You need to include a company logo in your website header, stored in the images folder.

Use Case:

- Linking external resources like CSS stylesheets.
- Embedding images and media files.
- Navigating between different pages within a website.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>File Paths</title>

</head>

<body>

  <h1>Welcome to Our Website</h1>

  
  <a href="pages/about.html"

</body>

</html>
```

Here in the above code we are giving the existing path of data in & <a> tag.

27) HTML Head

The <head> element contains metadata, links to stylesheets, scripts, and other important non-visible data that help in structuring and optimizing the webpage.

Scenario: You want to improve your webpage's SEO and define a title and meta description.

Use Case:

- Defining page titles and metadata for SEO.
- Linking external CSS and JavaScript files.
- Setting character encoding and responsive meta tags.

```
<!DOCTYPE html>

<html>

  <head>
    <title>My Website</title>
    <meta charset="UTF-8">
    <meta name="description" content="A sample webpage">
    <meta name="keywords" content="HTML, CSS, JavaScript">
    <meta name="author" content="John Doe">
    <link rel="stylesheet" href="styles.css">
  </head>

  <body>

    <h1>Welcome to My Website</h1>

  </body>

</html>
```

28) HTML Layout

HTML layout structures the content using semantic elements like <header>, <nav>, <section>, <article>, <aside>, and <footer>.

Scenario: You are building a structured webpage with navigation, a content area, and a footer.

Use Case:

- Creating structured web pages.
- Improving accessibility and SEO with semantic elements.
- Building responsive layouts using CSS.

```
<!DOCTYPE html><html><head>

  <title>HTML Layout</title>

  <link rel="stylesheet" href="layout.css">

</head><body>

  <header>

    <h1>Website Header</h1>

  </header>

  <nav>

    <ul>

      <li><a href="#">Home</a></li>

      <li><a href="#">About</a></li>

    </ul>

  </nav>

  <section>

    <h2>Main Content</h2>

    <p>This is the main section.</p>

  </section>

  <aside>

    <h2>Sidebar</h2>

    <p>Additional content.</p>

  </aside>

  <footer>

    <p>Footer Information</p>

  </footer>

</body></html>
```

29) HTML Responsiveness

Responsive design ensures a webpage adapts to different screen sizes using CSS media queries.

Scenario: You want your website to change background color when viewed on a mobile device.

Use Case:

- Optimizing websites for mobile and tablet users.
- Adjusting font sizes and layouts dynamically.
- Improving user experience on various devices.

```
<html>

<head>

  <title>Responsive Webpage</title>

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <style>

    body { font-family: Arial, sans-serif; }

    @media (max-width: 600px) {

      body { background-color: lightgray; }

    }

  </style>

</head>

<body>

  <h1>Responsive Design</h1>

  <p>Resize the window to see the effect.</p>

</body>

</html>
```

30) HTML Computer Codes

HTML provides special tags for displaying computer code, such as **<code>**, **<pre>**, **<kbd>**, **<samp>**, and **<var>**. These tags help in formatting code snippets and technical documentation.

Scenario: You want to share a JavaScript function on your blog in a readable format.

Use Case:

- Displaying programming code in blogs and documentation.
- Formatting inline code snippets in articles.
- Representing user input or system output.

```
<!DOCTYPE html>

<html>

<head>

  <title>Code Example</title>

</head>

<body>

  <h2>Inline Code:</h2>

  <p>Use <code>&lt;code>&gt;</code> for inline code.</p>

  <h2>Code Block:</h2>

  <pre>

    function greet() {

      console.log("Hello, World!");

    }

  </pre>

</body></html>
```

31) HTML Emoji

HTML supports the use of emojis using Unicode character codes or directly by typing the emoji characters. Emojis enhance user engagement and add visual appeal to web content. They are particularly useful in applications involving messaging, feedback, social interaction, and modern UI/UX designs.

Use Cases:

- Enhancing user messages in chat applications
- Improving emotional tone in content
- Gamifying feedback or rating systems

Practical:

```
<!DOCTYPE html>

<html>

<head>

  <title>Emoji Example</title>

</head>

<body>

  <p>&#128512; Smiley Emoji</p>

  <p>&#128151; Heart Emoji</p>

</body>

</html>
```

32) HTML Media

HTML supports embedding media elements like images, audio, and video directly into webpages. These elements allow multimedia content to be displayed natively in the browser without the need for plugins.

Use Case:

- Adding visual elements to blogs or articles.
- Embedding media for educational platforms.
- Displaying product images and promotional videos.

Practical: Adding image

```
<!DOCTYPE html>

<html>

<head>

  <title>HTML Media</title>

</head>

<body>

  <h2>Image Example</h2>

  

</body>

</html>
```

33) HTML Video

The <video> element allows the embedding of video content on webpages. It supports formats like MP4, WebM, and Ogg. Controls like play, pause, and volume are built-in, enhancing user interaction.

Use Case:

- Hosting video tutorials or webinars
- Embedding marketing or advertising videos
- Creating online learning and entertainment platforms

Practical:

```
<!DOCTYPE html><html>

<head><title>HTML Video</title></head>

<body>

  <video controls width="500">

    <source src="video.mp4" type="video/mp4">Your browser does not support the video tag.

  </video>

</body></html>
```

34) HTML Audio

The <audio> element is used to play sound files such as music, podcasts, or voice recordings. It supports common audio formats like MP3, Ogg, and WAV. Controls include play, pause, and volume.

Use Case:

- Integrating background music on webpages.
- Embedding audio lectures or language lessons.
- Streaming podcasts or audio commentary.

Practical:

```
<!DOCTYPE html>

<html>

<head>

  <title>HTML Audio</title>

</head>

<body>

  <audio controls>

    <source src="audio.mp3" type="audio/mpeg">

    Your browser does not support the audio element.

  </audio>

</body>

</html>
```

35) HTML Plug-ins

HTML supports embedding external content like PDFs or Flash using <object>, <embed>, and <iframe> .These tags are commonly used to display documents or media not natively supported by HTML.

Use Case:

- Displaying embedded PDF documents
- Integrating third-party widgets
- Running legacy applications in browsers

Practical:

```
<!DOCTYPE html> <html>

<head>

  <title>PDF Plugin</title>

</head>

<body>

  <object data="file.pdf" type="application/pdf" width="600" height="400"></object>

</body>

</html>
```

36) HTML YouTube

Using the **<iframe>** element, YouTube videos can be embedded into webpages. This enables integration of multimedia from YouTube's platform with full playback and control options.

Use Case:

- Sharing educational or promotional videos
- Embedding live streams or vlogs
- Enhancing content with video tutorials

Practical:

```
<!DOCTYPE html>

<html>

<head>

  <title>YouTube Embed</title>

</head>

<body>

  <iframe width="560" height="315" src="https://www.youtube.com/embed/example"
  frameborder="0" allowfullscreen></iframe>

</body></html>
```


37) HTML Geolocation

The Geolocation API allows web applications to retrieve geographical location information from the user's device. With the user's permission, developers can fetch latitude and longitude to provide location-based services.

Use Case:

- Mapping applications and store locators
- Personalizing content based on location
- Creating check-in or delivery tracking features

Practical:

```
<!DOCTYPE html>

<html>

<head>

  <title>Geolocation</title>

</head>

<body>

  <button onclick="getLocation()">Get Location</button>

  <p id="location"></p>

  <script>

    function getLocation() {

      navigator.geolocation.getCurrentPosition(function(position) {

        document.getElementById("location").innerHTML = "Latitude: " + position.coords.latitude +

          "<br>Longitude: " + position.coords.longitude;

      });

    }

  </script>

</body>

</html>
```

38) HTML Form

HTML forms collect user input using interactive controls like text fields, checkboxes, radio buttons, and submit buttons. Form data is typically submitted to a server for processing using **methods such as GET or POST**. Forms add the ability to web pages to not only provide the person viewing the document with dynamic information but also to obtain information from the person viewing it, and process the information. Include such things as:

- **Drop-down lists**
- **Check boxes**
- **Radio buttons**
- **Text boxes**

To insert a form we use the **<FORM>** tag. The rest of the form elements must be inserted in between the form tags.

Use Case:

- User registration, login, and authentication
- Collecting survey or feedback data
- Online orders and payment processing

<FORM> element attributes

- **ACTION:** is the URL of the CGI (Common Gateway Interface) program that is going to accept the data from the form, process it, and send a response back to the browser.
- **METHOD:** GET (default) or POST specifies which HTTP method will be used to send the form's contents to the web server. The CGI application should be written to accept the data from either method.
- **NAME:** is a form name used by VBScript or JavaScripts.
- **TARGET:** is the target frame where the response page will show up.

38.1.1 <INPUT type="text">

- **TYPE:** Type of INPUT entry field.
- **NAME:** Variable name passed to CGI application
- **VALUE:** The data associated with the variable name to be passed to the CGI application.
The default value is 28.
- **CHECKED:** Button/box checked
- **SIZE:** Number of visible characters in text field
- **MAXLENGTH:** Maximum number of characters accepted. **Default value 20 characters.**

Example:-

First Name: <INPUT TYPE="TEXT" NAME="FName" SIZE="15" MAXLENGTH="25">
--

38.1.2 <INPUT type="password">

pascode: <INPUT TYPE="PASSWORD" NAME="pwd" value="" SIZE="15" MAXLENGTH="25">

38.1.3 <INPUT type="Hidden">

Hidden: Used to send data to the CGI application that you don't want the web surfer to see, change or have to enter but is necessary for the application to process the form correctly. **Nothing is displayed in the browser.**

38.1.4 <INPUT type="Checkbox">

Check Box: Check boxes allow the users to select more than one option.

Checkboxes have the following attributes:

- **TYPE:** checkbox.
- **CHECKED:** is blank or CHECKED as the initial status.
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** is usually set to a value.

Example:-

Select country
jordan: <input type="CheckBox" Name="country" CHECKED>

Yemen: <input type="CheckBox" Name="country">

Qatar: <input type="CheckBox" Name="country">

38.1.5 <INPUT type="radio">

Radio Button: Radio buttons allow the users to select only one option.

Radio buttons have the following attributes:

- **TYPE:** radio.
- **CHECKED:** is blank or CHECKED as the initial status. Only one radio button can be checked
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** usually has a set value.

Example:-

```
jordan:<INPUT TYPE= "RADIO" CHECKED><BR>
Yemen<INPUT TYPE="RADIO "Name="country"><BR>
Qatar:<INPUT TYPE="RADIO" Name="country"><BR> <BR>
```

38.1.6 PUSH buttons:- <INPUT type="button">

Push Button: This element would be used with JavaScript to cause an action to take place.

Push Button has the following attributes:

- **TYPE:** button
- **NAME:** is the name of the button to be used in scripting.
- **VALUE:** determines the text label on the button.

```
<INPUT TYPE="button" VALUE="PressMe">
```

38.1.7 submit buttons:- <INPUT type="submit">

Submit: Every set of Form tags requires a Submit button. This is the element causes the browser to send the names and values of the other elements to the CGI Application specified by the ACTION attribute of the FORM element.

```
<INPUT TYPE="submit" VALUE="SubmitData " >
```

38.1.8 Reset button:- <INPUT type="reset">

Reset: It is a good idea to include one of these for each form where users are entering data. It allows the surfer to clear all the input in the form.

```
<INPUT TYPE="RESET" VALUE="Reset">
```

38.1.9 File:- <INPUT type="file">

File Upload: You can use a file upload to allow surfers to upload files to your web server.

```
<INPUT TYPE="File" name="myFile" size="30">
```

38.2 <TEXTAREA>

Textarea is an element that allows for free form text entry.

Textarea has the following attributes:

- **NAME:** is the name of the variable to be sent to the CGI application.
- **ROWS:** the number of rows to the textbox.
- **COLS:** the number of columns to the textbox.

```
<TEXTAREA COLS=40 ROWS=20 Name="comments" >
```

From observing the apathy of those about me during flag raising I concluded that patriotism if not actually on the decline is at least in a state of dormancy.

Written by Khaled Al-Fagih

```
</TEXTAREA>:
```

38.3 <SELECT>

The two following examples are <SELECT></SELECT> elements, where the attributes are set differently.

The Select elements attributes are:

- **NAME:** is the name of the variable to be sent to the CGI application.
- **SIZE:** this sets the number of visible choices.
- **MULTIPLE:** the presence of this attribute signifies that the user can make multiple selections. By default only one selection is allowed.

```
<BODY bgcolor=lightblue>
```

```
<form>
```

Select the cities you have visited:

```
<SELECT name="list" size=5>
```

```
<option>London</option>
```

```
<option>Tokyo</option>
```

```
<option> New York</option>
```

```
<option> LA</option>
```

```
</SELECT>
```

```
</form></BODY>
```

